Eenheid 2.2

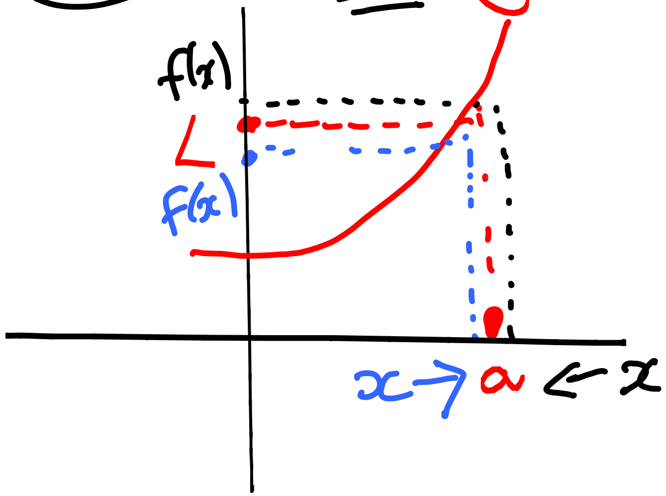
Note Title 2016/02/24

NB Die limiet van 'n funksie

Stewart bl. 83-94
Studiegids bl. 15

Definisie (informeel)

Veronderstel $f(x)$ is gedefinieer as x naby die getal a is.



dws. f is gedefinieer op 'n oop interval wat a bevat, maar $x \neq a$.

"Soos wat x-waardes nader kom na a maar nooit self a is nie, kom f(a) nader na L."

Ons skryf dit as volg: (limiet)

$$\lim_{x \to a} f(x) = L$$

Kyk bv. na $f(x) = \dfrac{e^{4x}-1}{x}$

$f(0) = \dfrac{e^{4(0)}-1}{0} = \dfrac{e^{0}-1}{0} = \dfrac{1-1}{0}$?? Ongedefinieerd

Wat gebeur met die funksie as ons na waardes naby 0 kyk?

x positief	$f(x)$
0.01	4.081077
0.001	4.0080167

x negatief	$f(x)$
−0.01	3.921056
−0.001	3.992010656

Hoe nader die waardes na 0 kom, van beide kante af (maar $x \neq 0$) hoe nader kom die waardes van $f(x)$ na 4. Dus $\lim_{x \to 0} \dfrac{e^{4x}-1}{x} = 4$

VOORBEELDE: Tegnieke om limiete te bepaal!

Probeer altyd eers die volgende metode

Direkte substitusie

① $\lim_{x \to -1} (x^2 + x - 2) = (-1)^2 + (-1) - 2 = -2$

② $\lim_{x \to 2} \frac{x^2 - 4}{x + 2} = \frac{(2)^2 - 4}{2+2} = \frac{0}{4} = 0$

As direkte substitusie nie werk nie

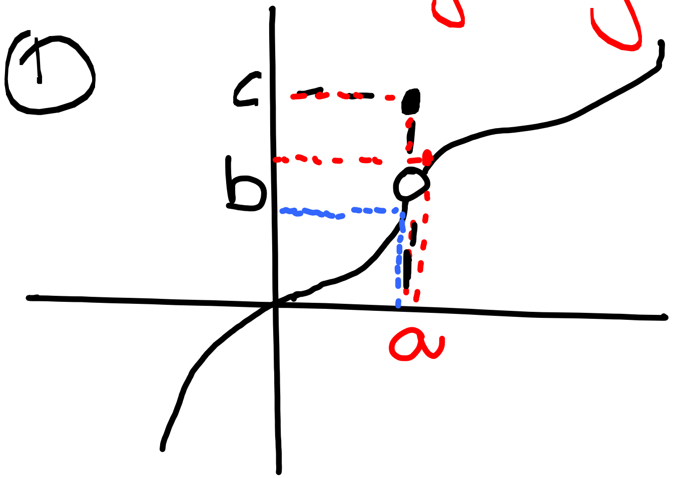
"Plan": ?? Faktoriseer !!

• ① $\lim_{x \to -3} \frac{x^2 - 9}{x + 3} \left(\frac{0}{0}\right) = \lim_{x \to -3} \frac{(x-3)\cancel{(x+3)}}{\cancel{x+3}} = -6$

Eensydige limiete (halwe limiete)

- Stuksgewys gedefineerde funksies.

①



- $f(a) = c$
- $\lim\limits_{x \to a^+} f(x) = b$ (limiet as x vanregs af al hoe nader aan a kom)

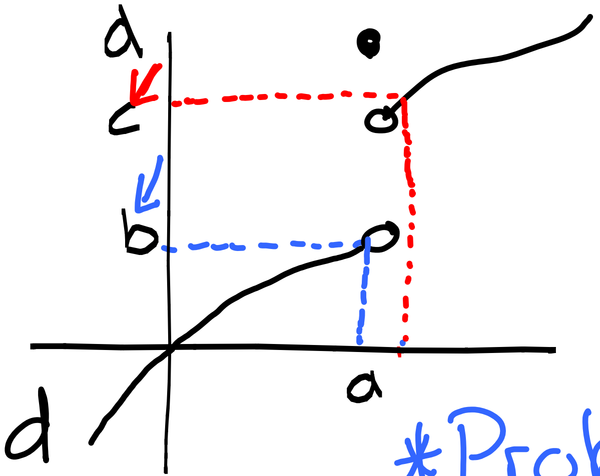
Regterlimiet

- $\lim\limits_{x \to a^-} f(x) = b$ (x kom al hoe nader na a van links af)

Linkerlimiet

Dan $\lim\limits_{x \to a} f(x) = b$

②



- $f(a) = d$

- $\lim\limits_{x \to a^+} f(x) = c$ (regterlimiet)
- $\lim\limits_{x \to a^-} f(x) = b$ (linkerlimiet)

*Probleem: $\lim\limits_{x \to a^+} f(x) \neq \lim\limits_{x \to a^-} f(x)$

Stelling:

$\lim_{x \to a} f(x) = ??$

$$\lim_{x \to a} f(x) = L \iff \lim_{x \to a^+} f(x) = L \text{ en } \lim_{x \to a^-} f(x) = L$$

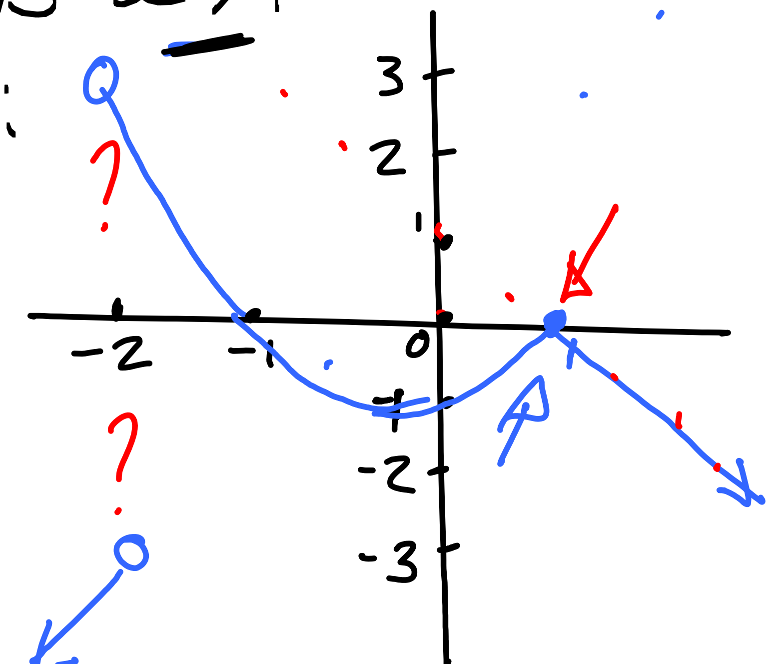
(bestaan)

Voorbeelde

① Gegee
$$f(x) = \begin{cases} 2x+1 & \text{as } x < -2 \\ x^2 - 1 & \text{as } -2 < x \leq 1 \\ 1 - x & \text{as } x > 1 \end{cases}$$

Vind, indien moontlik:

- $\lim_{x \to 10} f(x)$
- $\lim_{x \to 1} f(x)$
- $\lim_{x \to -2} f(x)$



- $\lim_{x \to 10} f(x)$ (as $x \to 10$ is $f(x) = 1-x$)

$= \lim_{x \to 10} (1-x) = 1 - 10 = -9$

- $\lim_{x \to 1} f(x) = ??$

$* \lim_{x \to 1^+} f(x) = \lim_{x \to 1^+} (1-x) = 0$

Yelling!!

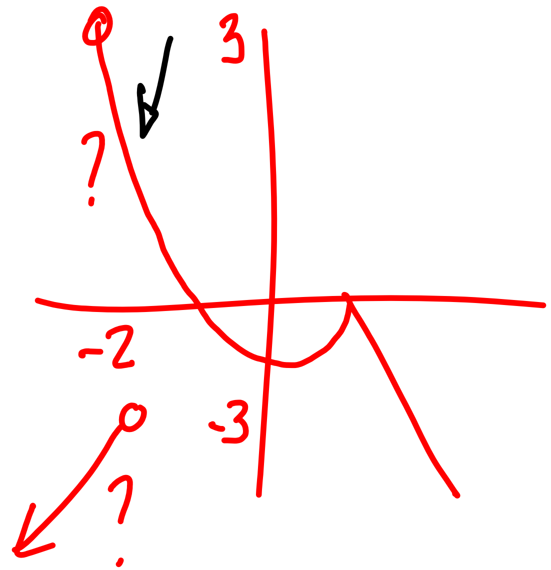
$* \lim_{x \to 1^-} f(x) = \lim_{x \to 1^-} (x^2-1) = 0$

$\checkmark$ Thus $\lim_{x \to 1} f(x) = 0$

- $\lim_{x \to -2} f(x) = ??$

$$\lim_{x \to -2^-} f(x) = \lim_{x \to -2^-} (2x+1) = -3$$

$$\lim_{x \to -2^+} f(x) = \lim_{x \to -2^+} (x^2-1) = 3$$



WANT: $\lim_{x \to -2^-} f(x) \neq \lim_{x \to -2^+} f(x)$

$\therefore \lim_{x \to -2} f(x)$ bestaan nie

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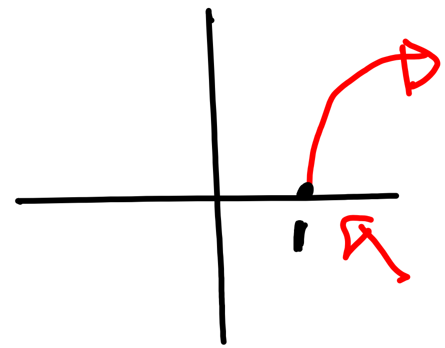
(2) $\lim_{x \to 1} \sqrt{x-1}$??



*Redenasie: $\lim_{x \to 1^+} \sqrt{x-1} = 0$ ✓

$x \to 1^+ \Rightarrow x > 1 \Rightarrow x-1 > 0$

* <u>Redenasie</u>: $\lim_{x\to 1^-}\sqrt{\underset{<0\,??}{x-1}}$ bestaan nie $x\to 1^- \Rightarrow x<1$
$\Rightarrow x-1<0$

<u>Gevolgtrekking</u>: $\lim_{x\to 1}\sqrt{x-1}$ bestaan nie

want $\sqrt{x-1}$ is ongedefinieerd as $x\to 1^-$

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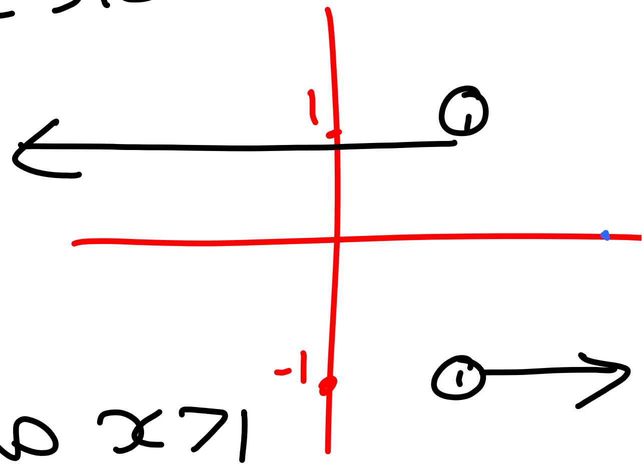
③ $f(x) = \dfrac{1-x}{|x-1|}$ • $\lim_{x\to 1} f(x) = ?$ • $\lim_{x\to 10} f(x) = ?$

$$f(x) = \begin{cases} \dfrac{1-x}{x-1} & \text{as } x-1>0 \\ \dfrac{1-x}{-(x-1)} & \text{as } x-1<0 \end{cases}$$

$$= \begin{cases} \dfrac{-(x-1)}{x-1} & \text{as } x>1 \\ \dfrac{1-x}{1-x} & \text{as } x<1 \end{cases} =^{*} \begin{cases} -1 & \text{as } x>1 \\ 1 & \text{as } x<1 \end{cases}$$



$$\lim_{x \to 1^+} f(x) = -1 \neq \lim_{x \to 1^-} f(x) = 1$$

Dus $\lim_{x \to 1} f(x)$ bestaan nie

- **Oneindige limiete** (antwoord is $\pm\infty$)

Informele definisie:

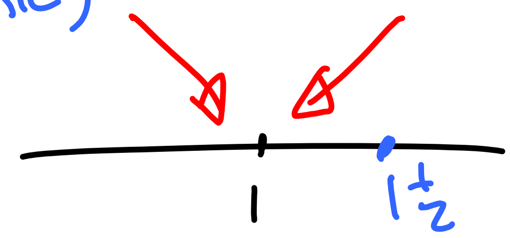
$\lim_{x \to a} f(x) = \infty$ Waardes van $f(x)$ kan willekeurig groot gemaak word (so groot as wat jy wil) deur x naby genoeg aan a te neem, maar $x \neq a$

Dus $\lim_{x \to a} f(x) = \infty$ beteken

Dus limiet bestaan nie!! Limiet bestaan net as limiet se antwoord 'n reële getal is.

Bv. $\lim\limits_{x\to 1} x+2 = 3$ (bestaan)

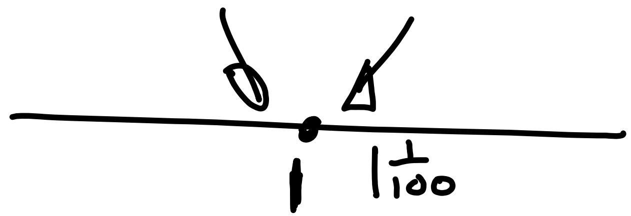
$\lim\limits_{x\to 1} \dfrac{1}{(x-1)^2} = +\infty$? (bestaan nie)



Een tipe oneindige limiete is in 'n vorm wat jy sal moet kan herken.

$$\frac{\text{nie-nul getal}}{\text{getal wat kleiner en kleiner word}} = ?$$

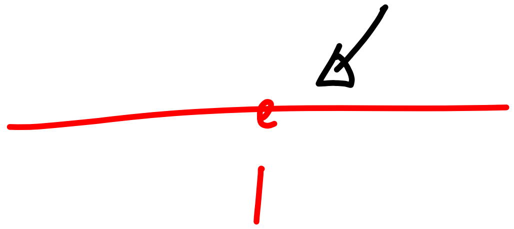
Bv $\lim\limits_{x\to 1} \dfrac{1}{(x-1)^2} = \infty$??



$$\begin{cases} x = 1\frac{1}{100} \Rightarrow x - 1 = \frac{1}{100} & f(x) = \frac{1}{\left(\frac{1}{100}\right)^2} = (100)^2 \\ x = \frac{99}{100} \Rightarrow x - 1 = -\frac{1}{100} & f(x) = \frac{1}{\left(-\frac{1}{100}\right)^2} = 100^2 \end{cases}$$

Dink

- $\lim\limits_{x \to 1^+} \frac{-4}{x-1} \frac{(-)}{(+)} \overset{?}{=} \boxed{-\infty}$



$x \to 1^+ \Rightarrow x > 1 \Rightarrow x - 1 > 0$

$\lim\limits_{x \to 1^-} \frac{-4}{x-1} \frac{-}{-} = +\infty$

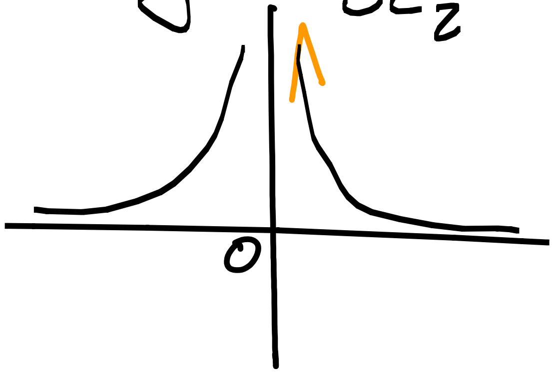
$x \to 1^- \Rightarrow x < 1 \Rightarrow x - 1 < 0$

- $\lim\limits_{x \to 1} \frac{-4}{x-1}$ bestaan nie

*Vertikale asimptote

• Spesiale vbd.

$$y = \frac{1}{x^2}$$



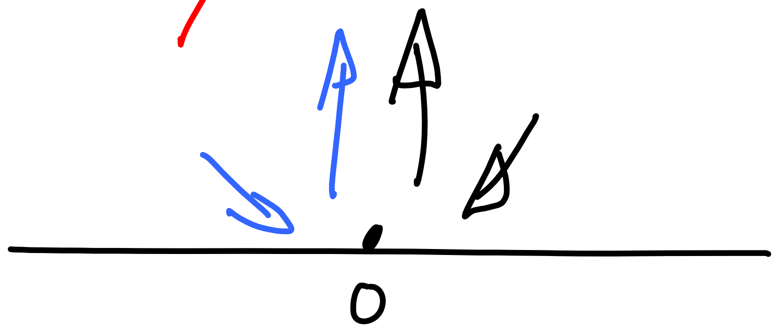
STAPPE:

• $D_f = \mathbb{R}\setminus\{0\} = (-\infty,0)\cup(0,\infty)$

• Is $\boxed{0} \in D_f$?

• Vermoed daar is 'n vertikale asimptoot by $x=0$.

• Moet dit toets/bevestig!!!



• Toets:

$$\begin{cases} \lim_{x\to 0^-} \frac{1}{x^2} = \infty \\ \lim_{x\to 0^+} \frac{1}{x^2} = \infty \end{cases}$$

Teorie:

$x = a$ is 'n Vertikale asimptoot van $y = f(x)$ as ten minste een van die volgende bewerings geld:

$$\lim_{x \to a} f(x) = \infty \qquad \lim_{x \to a} f(x) = -\infty$$

$$\lim_{x \to a^+} f(x) = \infty \qquad \lim_{x \to a^+} f(x) = -\infty$$

$$\lim_{x \to a^-} f(x) = \infty \qquad \lim_{x \to a^-} f(x) = -\infty$$

__Gevolgtrekking__: (vir gewone funksies!)

* Bepaal D_f.
* As $x = a$ in D_f is, dan is $x = a$ nie 'n vertikale asimptoot nie

* As $x = a$ nie in D_f is nie, dan kan $x = a$ MOONTLIK 'n VA wees.

* __Toets__ of $x = a$ 'n VA is of nie!!

By stuksgewys gedefineerde funksies kan daar uitsonderings wees.

VOORBEELDE:

① $f(x) = x+1$

$D_f = \mathbb{R}$ Geen vertikale asimptoot nie

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② • $f(x) = \dfrac{\sqrt{x}-3}{x-9}$ $x \geq 0$ $D_f = [0,9) \cup (9,\infty)$

$x-9 \neq 0 \Rightarrow x \neq 9$

Vermoed VA by $x = 9$

• Toets: $\displaystyle\lim_{x\to 9} \frac{\sqrt{x}-3}{x-9}$ $\left(\frac{0}{0}\right)$ NB

$$= \lim_{x\to 9} \frac{\cancel{\sqrt{x}-3}}{(\cancel{\sqrt{x}-3})(\sqrt{x}+3)} = \frac{1}{6}$$

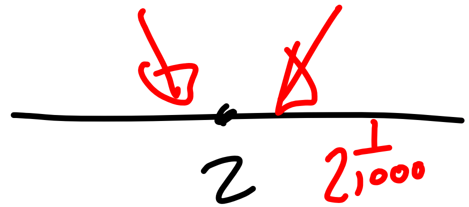
• Metode:
• Direk
• $\left(\frac{0}{0}\right)$ Faktoriseer
• Halwe limiete

?? Is $x=9$ 'n VA ??

Nee want $\displaystyle\lim_{x\to 9} f(x) \neq \infty$.

(3) $\lim\limits_{x \to 2} \dfrac{\overset{\neq 0}{1}}{(x-2)^2} = +\infty$

kleiner



- $D_f = \mathbb{R} \setminus \{2\}$
- VA by $\boxed{x = 2}$

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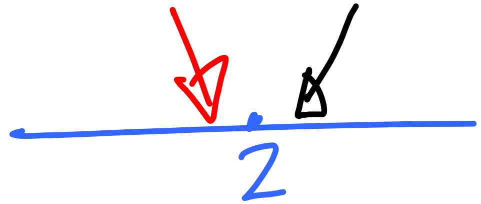
(4) $\lim\limits_{x \to 2^+} \dfrac{1}{x-2} = +\infty$

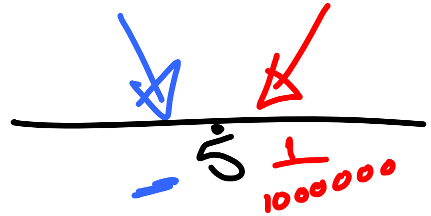
$\lim\limits_{x \to 2^-} \dfrac{1}{x-2} = -\infty$



Dus $\lim\limits_{x \to 2} f(x)$ bestaan nie

VA by $x = 2 \notin D_f$

⑤ $f(x) = \dfrac{2-x}{x-5}$ $\qquad \lim\limits_{x \to 5} f(x) = ?$



$$\lim_{x \to 5^+} \frac{\overset{(5)}{2-x}^{\,-}}{x-5^{\,+}} = -\infty$$

$$\lim_{x \to 5^-} \frac{\overset{(5)}{2-x}^{\,-}}{x-5^{\,-}} = +\infty$$

Dus $\lim\limits_{x \to 5} f(x)$ bestaan nie

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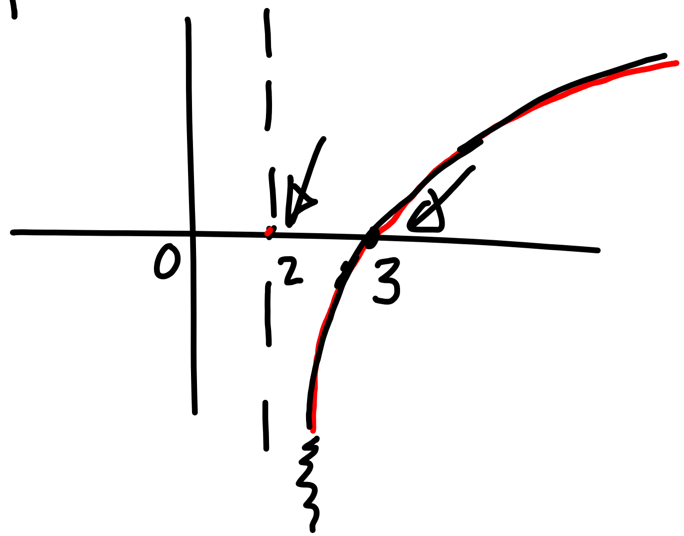
⑥ $f(x) = \dfrac{x^2-16}{x-4}$ $\qquad$ Is daar 'n VA? $\quad x=4$??

- $D_f = \mathbb{R} \setminus \{4\}$
- $\lim\limits_{x \to 4} \dfrac{x^2-16}{x-4} \left(\dfrac{0}{0}\right) = \lim\limits_{x \to 4} \dfrac{\cancel{(x-4)}(x+4)}{\cancel{x-4}} = 8$

Dus $x=4$ is nie 'n VA nie

⑦ $f(x) = \ln(x-2)$ $\qquad D_f = (2, \infty)$

$$\lim_{x \to 2^+} \ln(\overset{2}{x}-2) = -\infty$$



Dus $x = 2$ is 'n VA

- Onthou: $\ln(x-2) \leq 0$

kyk na: $x-2>0$ en $\Rightarrow$ $\ln(x-2) \leq 0$

>0 altyd

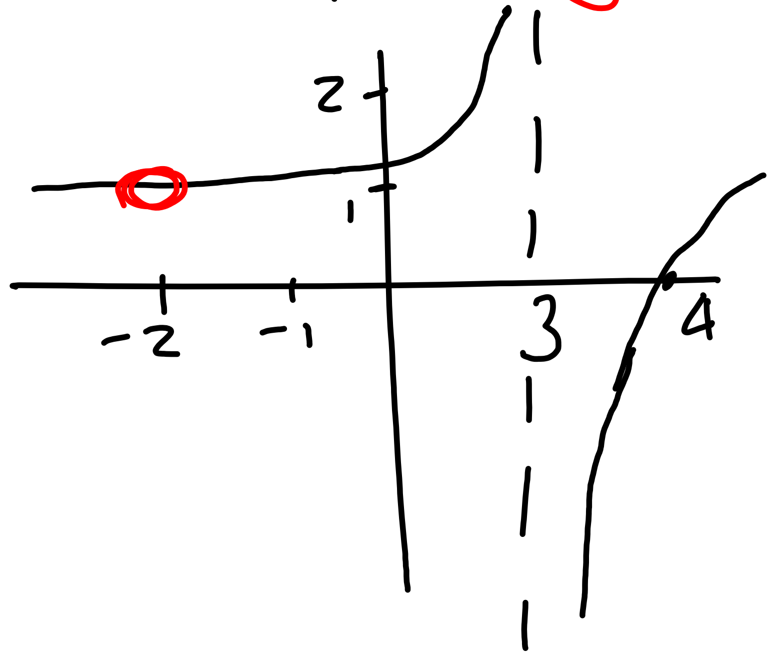
- Onthou: $e^x(1-x) > 0 \Rightarrow 1-x > 0 \Rightarrow$?

Onthou Tegniekbemeestering agter in Studiegids.

Maar: Wat van:

$$f(x) = \frac{x^2 - 2x - 8}{x^2 - x - 6} = \frac{(x-4)\cancel{(x+2)}}{(x-3)\cancel{(x+2)}}$$

• $D_f = \mathbb{R} \setminus \{-2, 3\}$



Toets net vir $x = 3$

$$\lim_{x \to 3^+} \frac{x-4}{x-3} \; \frac{(-)}{(+)} = -\infty$$

$$\lim_{x \to 3^-} \frac{x-4}{x-3} \; \frac{-}{-} = +\infty$$

Dus $x = 3$ is 'n VA